

P versus NP

The Barriers and the Wall

Diego Rincón

Independent

Abstract

The P versus NP problem asks whether every problem whose solution can be verified in polynomial time can also be solved in polynomial time. The answer is almost universally believed to be no: $\mathbf{P} \neq \mathbf{NP}$. But no proof exists. The situation is exceptional among the Millennium Problems: three independent proof barriers have been identified that rule out almost all known proof techniques. Relativization (Baker–Gill–Solovay) blocks oracle-relative arguments. Natural proofs (Razborov–Rudich) blocks combinatorial circuit lower bound methods under a cryptographic assumption. Algebrization (Aaronson–Wigderson) blocks algebraic extensions of relativization. What remains: a proof technique that is non-relativizing, non-naturalizing, and non-algebrizing. No such technique is known. The wall is three barriers thick.

1 The Problem

Definition 1 (Complexity classes). $\mathbf{P}$: languages decidable by a deterministic Turing machine in time $O(n^k)$ for some fixed k . $\mathbf{NP}$: languages where membership can be *verified* in polynomial time given a certificate (witness). Equivalently: languages decidable by a nondeterministic Turing machine in polynomial time.

Definition 2 (NP-completeness). A language L is $\mathbf{NP}$ -complete if: (1) $L \in \mathbf{NP}$, and (2) every $L' \in \mathbf{NP}$ is polynomial-time reducible to L . The first $\mathbf{NP}$ -complete problem: SAT (satisfiability of Boolean formulas), proved by Cook (1971) and independently Levin. NP-completeness is the structure of hardness: if any NP-complete problem is in $\mathbf{P}$, then $\mathbf{P} = \mathbf{NP}$.

Theorem 3 (Cook–Levin [1, 2]). *SAT is NP-complete. Therefore: $\mathbf{P} = \mathbf{NP}$ if and only if $\text{SAT} \in \mathbf{P}$.*

Definition 4 (The Prize question [3]). Is $\mathbf{P} = \mathbf{NP}$? Equivalently: does there exist a polynomial-time algorithm for SAT?

2 What Is Proved

Theorem 5 (Hierarchy theorems [4]). *Time and space hierarchy: $\mathbf{P} \subsetneq \mathbf{EXP}$, $\mathbf{PSPACE} \subsetneq \mathbf{EXPSPACE}$, $\mathbf{P} \subseteq \mathbf{PSPACE}$, $\mathbf{NP} \subseteq \mathbf{PSPACE}$. The diagonalization argument shows:*

more time (or space) strictly increases computational power. But diagonalization does not separate $\mathbf{P}$ from $\mathbf{NP}$ (see Relativization, below).

Theorem 6 (Circuit lower bounds for restricted models [5, 6, 7]). 1. AC^0 : *PARITY* requires exponential size (Fürst–Saxe–Sipser; sharp bounds by Håstad).

2. Monotone circuits: *CLIQUE* requires exponential size (Razborov 1985).

3. TC^0 lower bounds: no super-polynomial lower bounds known for general circuits computing any NP problem.

These results prove lower bounds for restricted circuit classes, but no lower bound of the form $SIZE(f) \geq n^{\omega(1)}$ for any explicit $f \in \mathbf{NP}$ in general Boolean circuits.

Remark 7 (The state of circuit complexity). A super-polynomial circuit lower bound for SAT would prove $\mathbf{P} \neq \mathbf{NP}$ (since $\mathbf{P} \subseteq \mathbf{P}/\text{poly}$, but $\text{SAT} \notin \mathbf{P}/\text{poly}$ would follow, giving $\text{SAT} \notin \mathbf{P}$). No such bound is known. The best known lower bound for SAT against general circuits is $3n - o(n)$ (a linear lower bound), far from super-polynomial.

3 The Three Barriers

Theorem 8 (Relativization [8]). *There exist oracles A, B such that:*

$$\mathbf{P}^A = \mathbf{NP}^A \quad \text{and} \quad \mathbf{P}^B \neq \mathbf{NP}^B.$$

Consequence: any proof of $\mathbf{P} \neq \mathbf{NP}$ (or $\mathbf{P} = \mathbf{NP}$) must use non-relativizing techniques — properties of the computation model that change with oracles.

Remark 9 (What relativization rules out). Most classical complexity theory arguments (diagonalization, simulation, padding) are relativizing. Baker, Gill, and Solovay showed in 1975 that no such argument can resolve $\mathbf{P}$ vs $\mathbf{NP}$: the answer depends on the internal structure of computation, not just the input-output behavior.

Theorem 10 (Natural proofs [9]). *A proof of circuit lower bounds is natural if it uses a property C of Boolean functions such that: (1) C is efficiently constructible (membership in C is decidable in $2^{O(n)}$), and (2) C is large (a $2^{-O(n)}$ fraction of functions satisfy C). Under the assumption that strong pseudorandom generators exist (a widely believed cryptographic hypothesis): natural proofs cannot prove super-polynomial circuit lower bounds for any $f \in \mathbf{NP}$. Consequence: most combinatorial circuit lower bound arguments are natural, hence blocked by the cryptographic assumption.*

Remark 11 (What natural proofs rules out). The Razborov–Rudich barrier explains why methods that worked for restricted circuits (AC^0 , monotone circuits) do not extend to general circuits: those methods are natural. A proof of $\mathbf{P} \neq \mathbf{NP}$ must either use a non-natural proof or contradict the existence of pseudorandom generators.

Theorem 12 (Algebrization [10]). *Define an algebraic oracle: an oracle A together with a low-degree extension $\tilde{A}$ over a field. An argument algebrizes if it relativizes with respect to algebraic oracles. There exist algebraic oracles A, B such that $\mathbf{P}^A = \mathbf{NP}^A$ and $\mathbf{P}^B \neq \mathbf{NP}^B$ with respect to algebraic extensions. Consequence: algebraic arguments (including interactive proofs, the PCP theorem proof, $IP = PSPACE$) do not resolve $\mathbf{P}$ vs $\mathbf{NP}$ either.*

Remark 13 (The three barriers together). Relativization rules out diagonalization. Natural proofs rule out combinatorial circuit lower bounds. Algebrization rules out algebraic extensions of relativization. The known techniques that have proved all major complexity results fall under one of these three categories. A proof of P vs NP must lie entirely outside all three. No such technique has been identified.

4 Three Forms of the Wall

Theorem 14 (The wall in three languages). *The following are equivalent, each a formulation of what must be proved:*

1. **Algorithmic:** *No polynomial-time algorithm exists for SAT (or any NP-complete problem).*
2. **Circuit:** *SAT requires super-polynomial Boolean circuits: $\text{SIZE}(\text{SAT}_n) \geq n^{\omega(1)}$.*
3. **Meta-mathematical:** *A non-relativizing, non-naturalizing, non-algebrizing argument establishes that no polynomial-time simulation of nondeterminism exists.*

Each is equivalent to $P \neq NP$. Proving any one unconditionally proves all three and solves the Prize.

5 State of the Art

Theorem 15 (What is proved). *The following hold unconditionally:*

1. $P \subsetneq \text{EXP}$ (time hierarchy).
2. SAT is NP-complete (Cook–Levin).
3. Thousands of NP-complete problems are catalogued.¹
4. $\text{PARITY} \notin AC^0$: constant-depth lower bounds (Håstad).
5. $\text{CLIQUE} \notin \text{monotone-}P$: monotone circuit lower bounds (Razborov).
6. The three barriers rule out virtually all known proof techniques.
7. $P \neq NP$ is believed by essentially all practitioners.

The following are equivalent to the Prize (not proved):

1. $\text{SAT} \notin P$ (or any NP-complete problem).

¹The reach of NP-hardness extends to recreation. Optimal play in Pokémon—minimizing turns to defeat an arbitrary opponent team—is an instance of adversarial planning with exponential branching, studied in the NP-hard regime [12]. At scale: Pokémon GO [11] deploys this structure globally. Millions of players simultaneously optimize routes through PokéStop graphs—a distributed approximation of TSP running on planetary infrastructure. The difficulty of the Prize is not purely academic.

2. Any super-polynomial circuit lower bound for SAT in general circuits.
3. Any non-relativizing, non-naturalizing, non-algebrizing argument distinguishing deterministic from nondeterministic polynomial time.

Remark 16 (The wall). The P vs NP wall is different from the other Millennium walls. For RH, NS, YM, Hodge, BSD: the wall is a single named open step at the boundary of what the current formalism can reach. For P vs NP: the wall is meta-mathematical. It is not that we cannot execute the last step; it is that we do not know what kind of step to take. Three barriers have eliminated the known directions. What remains is the void where the proof must live. We do not know what needs to be proved. We know that we do not know.

References

- [1] Cook, S. A. (1971). The complexity of theorem-proving procedures. *Proc. 3rd Annual ACM STOC*, 151–158.
- [2] Levin, L. A. (1973). Universal search problems. *Problems of Information Transmission*, 9(3), 265–266.
- [3] Cook, S. A. (2000). P versus NP. *Clay Mathematics Institute Millennium Problems*.
- [4] Hartmanis, J. and Stearns, R. E. (1965). On the computational complexity of algorithms. *Transactions of the American Mathematical Society*, 117, 285–306.
- [5] Fürst, M., Saxe, J. B., and Sipser, M. (1984). Parity, circuits, and the polynomial-time hierarchy. *Mathematical Systems Theory*, 17(1), 13–27.
- [6] Håstad, J. (1987). *Computational Limitations of Small-Depth Circuits*. MIT Press.
- [7] Razborov, A. A. (1985). Lower bounds on the monotone complexity of some Boolean functions. *Soviet Mathematics Doklady*, 31, 354–357.
- [8] Baker, T., Gill, J., and Solovay, R. (1975). Relativizations of the $P = ? NP$ question. *SIAM Journal on Computing*, 4(4), 431–442.
- [9] Razborov, A. A. and Rudich, S. (1997). Natural proofs. *Journal of Computer and System Sciences*, 55(1), 24–35.
- [10] Aaronson, S. and Wigderson, A. (2009). Algebrization: a new barrier in complexity theory. *ACM Transactions on Computation Theory*, 1(1), 1–54.
- [11] Niantic, Inc. (2016). *Pokémon GO* [Mobile application]. San Francisco, CA. niantic.com/games/pokemon-go.
- [12] Aloupis, G., Demaine, E. D., Guo, A., and Viglietta, G. (2014). Classic Nintendo games are (computationally) hard. *International Conference on Fun with Algorithms (FUN 2014)*, LNCS 8496, Springer. arXiv:1203.1895.
- [13] Rincón, D. (2026). P versus NP: Where the Adaptive Template Breaks. Preprint.